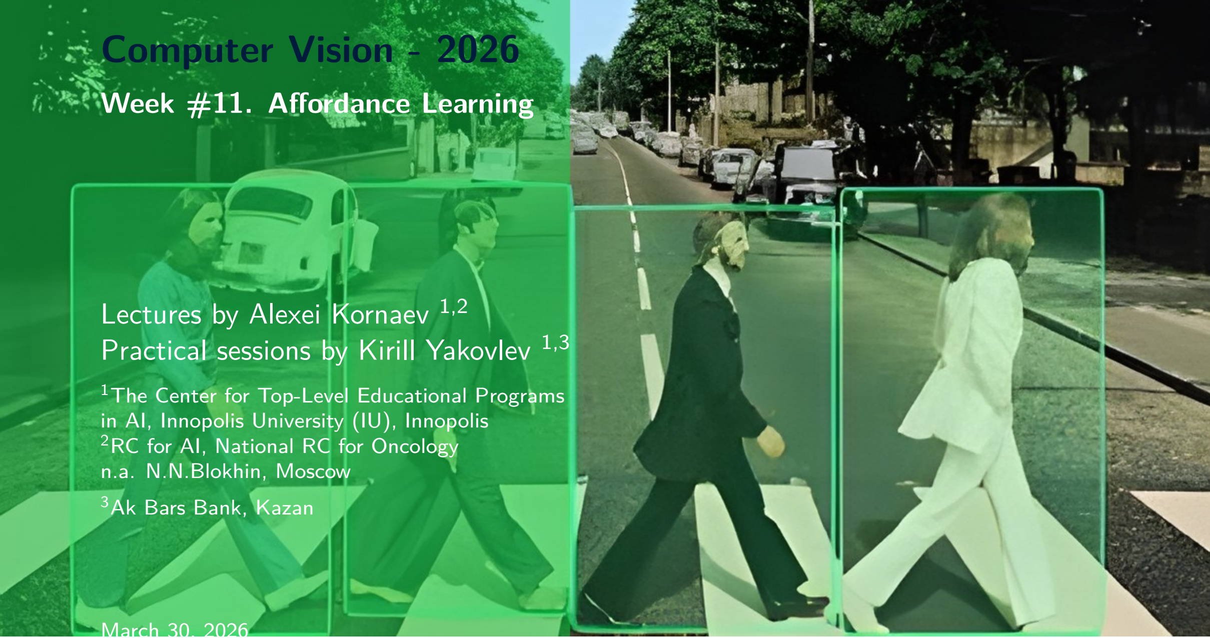

Computer Vision - 2026

Week #11. Affordance Learning

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Agenda

Affordance

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Perception Is Not the Goal

Traditional computer vision focused on recognition:

$$\text{Image} \rightarrow \text{Label}$$

Modern embodied systems require:

$$\text{Scene} \rightarrow \text{Action}$$

The objective shifts from understanding objects to predicting consequences.

- perception describes the world
- action changes the world
- intelligence requires predicting the result of actions

From Recognition to Interaction

Recognizing an object does not determine how it can be used.

Example:

- a chair can be:
  - sat on
  - lifted
  - pushed
  - stacked

The key question becomes:

What can I do with this object?

This question defines the concept of **affordance**.

Affordance Requires Physical Reasoning

The system must predict physical outcomes before acting.

Example reasoning:

- Is the structure stable?
- Will the object fall?
- Will the action cause damage?
- How much force is required?

Affordance prediction therefore requires:

- geometry
- dynamics
- constraints
- causal reasoning

Affordance Inference Pipeline

Step	Operation
1	Capture visual observation
2	Estimate scene geometry
3	Identify candidate interaction regions
4	Evaluate physical feasibility
5	Select stable action
6	Execute command

The decision rule can be expressed as:

$$a^* = \arg\max_a \; P(a \mid s)$$

The model selects the action with the highest probability of success.

Learning Outcomes

After completing this lecture, students should be able to:

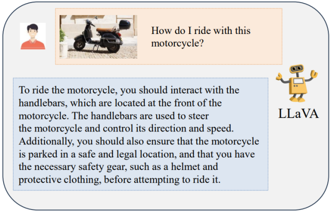
- **Explain** the concept of **affordance** as the link between perception and action in modern vision-language-action (VLA) systems
- **Distinguish** between **implicit** and **explicit** affordance representations in contemporary robotics architectures
- **Analyze** failure cases of foundation models under distribution shift (e.g., object pose change, geometry variation)
- **Interpret** the role of **physical constraints** (support region, center of mass, stability) in decision-making for embodied agents
- **Evaluate** model behavior using experimental evidence from a simulated stability task
- **Critically assess** whether a vision-language model demonstrates reasoning or pattern memorization

Core message of the lecture

Modern AI systems can recognize and act, but reliable behavior in the physical world requires structured reasoning about geometry, interaction, and physics.

Section 1. Modern Affordance Models

Affordance Grounding in the Wild



Task

Given: an image I, an action-aware text query T predict a dense affordance map M:

$$(I, T) \to M$$

Example

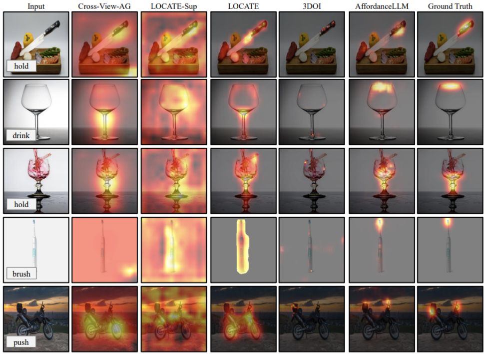
"What part of the motorcycle should we interact with in order to ride it?"

Key difficulty

Affordance requires more than appearance: object parts, 3D geometry, functionality, human-object interaction priors, commonsense reasoning,

AffordanceLLM formulates affordance grounding as language-conditioned dense prediction Qian et al. [2024]. Paper: CVPRW 2024 PDF.

Why Use a Vision-Language Model for Affordance?



Motivation

Large vision-language models already encode useful world knowledge about:

- object functionality
- part-action relations
- typical human-object interactions
- language-conditioned reasoning

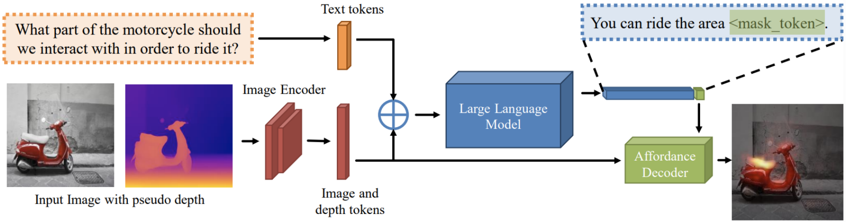
Paper intuition

If a VLM can answer **how** to use an object, then its internal representation may help localize **where** interaction should happen.

Shift in viewpoint

Affordance is not only a pixel-labeling problem. It becomes a **reasoning problem conditioned by language**.

AffordanceLLM: Overview



Core idea

RGB image, pseudo-depth, and text prompt are processed jointly by a multimodal LLM. The LLM is trained to emit a special `<mask_token>`; its hidden state is used as a query for an affordance decoder that predicts the final affordance map.

$$\text{image} + \text{depth} + \text{text} \longrightarrow \texttt{<mask\_token>} \longrightarrow \text{affordance map}$$

Figure 3 from Qian et al. [2024]. Paper: AffordanceLLM.

Data, Splits, and Evaluation

Dataset

Main benchmark: **AGD20K**. Training uses only images with **dense affordance annotations**.

Split	Train	Test
Easy (dense)	1135	540
Hard (dense)	868	807

Generalization setup

The central evaluation goal is generalization to **unseen object categories**.

- **Easy split**: original unseen split
- **Hard split**: train/test categories separated with lower semantic overlap

Why the split matters

This paper is not mainly about same-category performance. It tests whether language-conditioned representations help **in-the-wild affordance transfer**.

Metrics:

KLD ↓, SIM ↑, NSS ↑

Dataset and split details follow Qian et al. [2024]. AGD20K project page: AGD20K.

Architecture: Putting the Puzzle Together

Inputs

- RGB image I
- pseudo-depth map D
- text prompt T

Prompt template:

```
What part of the <object> should we interact with

in order to <action> it?
```

Visual side

- image encoder: **OWL-ViT**
- higher-resolution visual grounding than standard CLIP-ViT
- output: 576 tokens of dimension 768
- projected to 1024
- neighboring tokens concatenated to reduce memory

Language side

- multimodal backbone: **LLaVA-7B**
- language model: **LLaMA-7B**
- text tokens + visual tokens processed jointly

Affordance mechanism

- LLM predicts a special <mask_token>
- hidden state of <mask_token> becomes query q
- decoder uses q with image features to predict map

$$M = \mathrm{Decoder}(F_I, q)$$

Decoder

Lightweight decoder inspired by SAM/3DOI, with an additional transposed convolution layer for better output resolution.

$$A, M = \mathrm{AffordanceLLM}(F_I, F_D, F_T)$$

Objectives

Joint objective

The model is trained with a dense affordance loss and a language-model loss:

$$L = L_{\text{aff}} + \lambda L_{\text{text}}$$

where

- L_{aff}: binary focal loss on affordance map
- L_{text}: text generation cross-entropy
- $\lambda = 0.01$

Why focal loss?

Positive affordance pixels are sparse. The paper uses asymmetric weights:

- positive: 0.95
- negative: 0.05

to emphasize informative foreground regions.

Interpretation

The text branch encourages the multimodal LLM to preserve language-conditioned reasoning. The affordance branch forces the model to convert that reasoning into a dense spatial prediction.

Conceptual view

reasoning token $\longrightarrow$ grounded interaction map

objective from Qian et al. [2024]. This is the central link between language supervision and dense affordance grounding.

Training Algorithm

Algorithm 1: AffordanceLLM training pipeline (lecture reconstruction from the paper)

Input: RGB image I, affordance label Y, object name o, action name a
Output: Predicted affordance map M

Generate pseudo-depth map $D \leftarrow \mathrm{DPT}(I)$
Construct prompt T: `"What part of the` o `should we interact with in order to` a `it?"`
Extract RGB features $F_I \leftarrow \mathrm{OWL\text{-}ViT}(I)$
Extract depth features $F_D \leftarrow \mathrm{OWL\text{-}ViT}(D)$
Project visual tokens to language dimension
Tokenize text and concatenate multimodal tokens
Run multimodal LLM (LLaVA/LLaMA backbone)
Obtain hidden state of special <mask_token> and project to query q
Decode affordance map:

$$M \leftarrow \mathrm{Decoder}(F_I, q)$$

Compute affordance loss $L_{\text{aff}}(M, Y)$ using binary focal loss
Compute text loss L_{text} using language-model cross-entropy
Compute total loss:

$$L = L_{\text{aff}} + \lambda L_{\text{text}}$$

Update trainable parameters by backpropagation

Implementation notes

er reports PyTorch + HuggingFace, 8×A100 40GB, FSDP, batch size 4, learning rate $2 \cdot 10^{-5}$ Qian et al. [2024].

Results on AGD20K

Method	KLD ↓	SIM ↑	NSS ↑
LOCATE	1.985	0.348	0.787
LOCATE-Sup	1.872	0.350	0.801
Cross-View-AG	2.385	0.310	0.665
Cross-View-AG+	1.896	0.343	0.793
3DOI	1.805	0.358	0.905
AffordanceLLM	**1.661**	**0.361**	**0.947**

What matters most

The strongest advantage appears on the **hard split**, where the model must generalize to less similar unseen categories. This is exactly where language-conditioned world knowledge and 3D cues should help.

Qualitative pattern

The paper reports that:

- LOCATE often spreads activation across too much of the object
- 3DOI often predicts too small a region
- AffordanceLLM tends to produce the most plausible interaction map

summarized from Qian et al. [2024]. You may add the paper's Figure 4 here if space permits.

Ablations and Main Takeaways

Important ablations

- Full natural-language prompt $>$ weaker prompt variants
- OWL-ViT $>$ CLIP-ViT for this grounding task
- Pseudo-depth improves performance
- Full system performs best on the hard split

Why these results make sense

- language prompt injects richer semantic context
- OWL-ViT improves localization
- depth supplies missing 3D structure

Main conclusion

AffordanceLLM shows that affordance grounding benefits from combining:

- language-conditioned world knowledge
- visual grounding
- 3D cues
- dense spatial decoding

Bridge to modern VLA systems

This is not yet a full policy model, but it already has the key ingredients of modern VLA systems:

$$\text{perception} + \text{language} + \text{knowledge} \rightarrow \text{action prediction}$$

Later VLA models push the same idea further:

$$\text{affordance map} \rightarrow \text{action token} \rightarrow \text{control policy}$$

INNOPOLIS UNIVERSITY s and discussion from Qian et al. [2024]. This section naturally connects to VLA papers such as RT-2 and OpenVLA [illegible] et al. [2023]; Kim et al. [2024].

Paper Reading: AffordanceLLM

AffordanceLLM: Grounding Affordance from Vision Language Models

This work demonstrates that vision-language models can infer interaction regions directly from visual observations [Qian et al., 2024].

Key idea:

- large pretrained models encode functional knowledge about objects
- affordances can be predicted without explicit physics simulation
- reasoning emerges from large-scale training

Core implication:

affordance becomes a learned capability

Modern VLA Models: Affordance Implementations

Approach	Affordance Representation	Key Models
Implicit (pretrained backbone)	Distributed across attention layers	RT-2 Brohan et al. [2023], OpenVLA Kim et al. [2024]
Dense heatmap	Pixel-level interaction score map	RoboAfford Chen et al. [2023], AffCorrs Hadjivelichkov et al. [202
Keypoint / contact prediction	2-D / 3-D contact points	RoboPoint Yuan et al. [2024], UniAff Xu et al. [2024]
LLM chain-of-thought	Natural-language affordance reasoning	SayCan Ahn et al. [2022], CaP Liang et al. [2023]
3-D contact fields	Point-cloud / voxel interaction volume	3D-VLA Zhen et al. [2024], RoboUniView Liu et al. [2024]
World-model rollout	Predicted outcome quality	UniSim Yang et al. [2024]

Table: Taxonomy of affordance implementations in modern VLA systems.

Trend

Approaches are converging on **hybrid pipelines**: a VLM reasons about **what** affords interaction, a spatial module predicts **where**, and a diffusion / flow-matching policy determines **how** to execute the action.

Affordance in VLAs: Is It Solved? (2025–2026)

Empirical finding

End-to-end VLAs (**e.g.** π_0, OpenVLA) **memorise spatial patterns** rather than reasoning about affordance: success drops to $\sim$15 % after a 10 cm object shift Zheng et al. [2025]. Implicit encoding is **necessary** but **not sufficient**.

Open Problem	Status	Representative Work
Affordance under object pose shifts	Active — spatial modules needed	SAF Zheng et al. [2025]
Long-horizon affordance chaining	Active — CoT still unreliable	CoA-VLA Chen et al. [2025], VLA-R1 Shen et al. [2025]
Physical / contact-rich affordance	Very open	π_0 Black et al. [2024]
Cross-embodiment transfer	Active — architecture-level challenge	InternVLA-M1 Team [2025]
Affordance from zero / few demos	Open — promising early results	UAD Ye et al. [2025]

Table: Affordance research gaps identified in the 2025–2026 VLA literature.

2025 consensus Liu et al. [2025]

The field decisively shifted toward **explicit** affordance modelling (geometry, spatial constraints, chain-of-affordance reasoning) because implicit weight-space representations fail under real-world distribution shifts.

Section 2. Physical Reasoning in Embodied AI

From Affordance to Physical Reasoning

Modern embodied systems are converging toward a layered architecture:

$$\text{Perception} \rightarrow \text{Affordance} \rightarrow \text{Reasoning} \rightarrow \text{Control}$$

Motivation

Vision-language models can recognize objects and predict actions, but they often fail to anticipate physical consequences. Examples:

- placing a heavy object on a fragile surface
- stacking unstable structures
- predicting motion after contact

Cosmos-Reason1 introduces a dedicated reasoning model to bridge perception and action Research [2025].

Core Idea of Cosmos-Reason1

System Overview

Input video is processed by a vision encoder, projected into language tokens, and reasoned over by a multimodal language model. The system generates:

- physical explanations
- predictions
- next-step decisions

$$\text{Video} \rightarrow \text{Understanding} \rightarrow \text{Reasoning} \rightarrow \text{Decision}$$

Cosmos-Reason1 perceives the physical world through video input and generates decisions using long chain-of-thought reasoning processes Research [2025].

Ontology of Physical Intelligence

Physical Common Sense Ontology

The model organizes knowledge about the physical world into structured categories: Space, Time, Fundamental Physics
Each category contains fine-grained reasoning tasks such as:

- collision prediction
- object stability
- motion direction
- temporal ordering

Embodied Reasoning Ontology

Defines reasoning abilities for different agents:

- human
- robot arm
- humanoid
- autonomous vehicle

INNOPOLIS UNIVERSITY er introduces hierarchical ontologies to define and measure physical reasoning capabilities Research [2025].

Training Pipeline for Physical Reasoning

Stage	Operation
1	Pretrain multimodal vision-language model
2	Curate physical reasoning dataset
3	Supervised fine-tuning on physical tasks
4	Reinforcement learning optimization
5	Evaluate reasoning performance

Pretrained VLM $\rightarrow$ Physical AI SFT $\rightarrow$ Physical AI RL

Training Data

Dataset contains approximately:

- video-text pairs
- reasoning traces
- physical common sense questions

The model is trained using supervised fine-tuning followed by reinforcement learning on physical reasoning tasks Research [2025].

Results and Implications for Embodied AI

Model	Physical Reasoning Score
GPT-4o	55.6
OpenAI o1	59.9
Cosmos-Reason1-56B	60.2

Key Findings

- Physical reasoning improves significantly after reinforcement learning
- Structured ontologies help define measurable capabilities
- Reasoning models enable safer and more reliable robot decisions

Connection to Modern VLAs

Affordance prediction determines where to act.
Reasoning determines whether the action is safe.
Future embodied systems will integrate:

$$\text{Affordance} + \text{Reasoning} + \text{Control}$$

Cosmos-Reason1 demonstrates improved performance on physical common sense and embodied reasoning benchmarks Research [2025]

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